

AAYUSH VATS

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SKILLS

Languages: Java, JavaScript, TypeScript

Frontend & Backend: React.js, Next.js, Three.js, Node.js, Express.js, REST APIs, Prisma ORM

Databases: MongoDB, MySQL, Redis

DevOps, Cloud & Tools: Docker, AWS, Git, GitHub Actions, Linux

Core CS: Data Structures & Algorithms, Operating Systems, Computer Networks, DBMS

EXPERIENCE

Software Development Engineer Intern

Dec 2025 – Mar 2026

BitwiseLearn | Node.js, Express.js, MongoDB, Piston API

Remote

- Architected a backend infrastructure for a **multi-tenant e-learning platform** utilizing Role-Based Access Control (RBAC) to support secure access and improve scalability across five distinct user roles.
- Developed 40+ REST APIs** for course management, enrollment workflows, assessments, and learning operations using Node.js, Express.js, and MongoDB.
- Implemented a sandboxed code execution engine using the Piston API, enabling secure compilation and runtime testing across **17 programming languages**.

PROJECTS

Epiphany – Learning Management System 🐙 | Next.js, Express.js, Bun, JWT, MongoDB

- Engineered a full-stack Learning Management System (LMS) with **5 RBAC roles** supporting course management, enrollments, assignments, assessments, coding practice, and progress tracking.
- Designed a proctored assessment platform supporting **2 assessment formats (coding and MCQs)** with automated evaluation, JWT-based authentication, and detailed performance reports.
- Built a **LeetCode-inspired coding practice environment** featuring real-time code execution, custom test-case validation, submission history, and instant feedback.

Commit-AI – AI-Powered Git Workflow Assistant (npm) | TypeScript, Node.js, OpenAI, Octokit

- Engineered an AI-powered CLI featuring **7 Git automation commands** that streamline repository management, intelligent commit generation, branch creation, and code review workflows.
- Automated 5 repository operations** by integrating simple-git, Octokit, and the GitHub REST API, enabling one-command version control, remote synchronization, and automated PR generation.
- Implemented support for **3 interchangeable LLM providers (OpenAI, Gemini, and OpenRouter)**, enabling AI-generated conventional commits, branch names, pull request titles, and descriptions.

Trackio – Collaborative Progress Tracking Platform 🐙 | React, JavaScript, Tailwind CSS

- Built a collaborative platform** that enables users to create modular groups, tracking learning progress and coding streaks collectively.
- Programmed 5+ analytics modules**, including leaderboards, progress charts, streak tracking, dashboards, and activity insights.
- Embedded an AI assistant to provide contextual guidance and platform-wide user support, streamlining user onboarding and daily troubleshooting.

EDUCATION

Bachelor of Engineering in Computer Science

Aug 2024 – Jun 2028

Chitkara University | Punjab, India

– 9.49 CGPA

ACHIEVEMENTS

- Leadership:** Served as the Head of DSA & Programming at Open Source Chandigarh, mentoring 10+ students in advanced algorithms and competitive programming techniques.
- Competitive Programming:** Solved over 500 Data Structures and Algorithms (DSA) problems across leading platforms, including LeetCode, CodeChef, and GeeksforGeeks.
- Open Source:** Contributed actively to open-source projects, earning selection for participation in Hacktoberfest and the GirlScript Summer of Code.